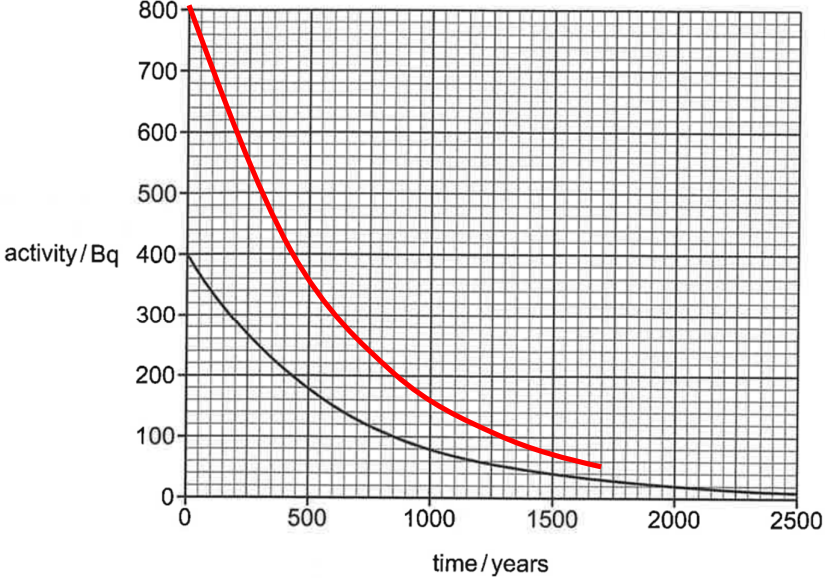


|                 |                                                                                                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6(a)(i)</b>  | <p>Radioactive means that the nuclei of the isotope are unstable and undergo random and spontaneous decay.</p> <p>The radiation emitted is in the form of alpha particles or beta particles or gamma rays.</p>                                                                                        |
| <b>6(a)(ii)</b> | <p>Half-life is the average time taken for the number of nuclei or mass or activity of the isotope to reduce to half its initial value.</p>                                                                                                                                                           |
| <b>6(b)(i)</b>  | <p>From Fig. 6.1</p> <ul style="list-style-type: none"> <li>- time taken for activity to decrease from 400 Bq to 200 Bq is 425 years</li> <li>- time taken for activity to decrease from 200 Bq to 100 Bq is 425 years</li> </ul> <p>Therefore half-life is 425 years.</p>                            |
| <b>6(b)(ii)</b> |  <p style="text-align: center;"><b>Fig. 6.2</b></p>                                                                                                                                                                |
| <b>6(c)</b>     | <p>The activity of 4 g of bone from a living human</p> $= \frac{270}{3} = 90 \text{ Bq}$ <p>Since the activity of 4 g of bone from an ancient human skeleton is 45 Bq, which is half of 90 Bq, the age of the ancient human skeleton is equal to the half-life of carbon-14, which is 5700 years.</p> |
| <b>7(a)</b>     | <p>The resistance of a circuit component is the ratio of the potential difference</p>                                                                                                                                                                                                                 |